# Hochschild Homology of the b-Pseudodifferential Symbol Algebra

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**Abstract**

Let $X$ be a compact $n$-manifold with embedded boundary and let $E \to X$ be a complex vector bundle. We compute the continuous Hochschild homology of the Benameur–Nistor Laurent complete-symbol algebra of the b-groupoid:

$$\mathcal{A}_{b,L}(X;E) = \mathcal{O}(X)\left(\Psi^{\infty}(\mathcal{G}_b;E)/\Psi^{-\infty}(\mathcal{G}_b;E)\right).$$

Here $\mathcal{O}(X)$ consists of functions with finite Laurent singularities at the boundary, and the algebra carries the published topologically filtered completion. With completed continuous Hochschild chains,

$$\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X;E)) \cong H_L^{2n-k}({}^bS^*X \times S^1_\sigma;\mathbb{C}).$$

The circle $S^1_\sigma$ records symbol order and $H_L^*$ is Laurent de Rham cohomology at the boundary. The displayed vector-space isomorphism is noncanonical; the filtration and its associated graded are natural. We also give two alternative reconstructions: a Čech–de Rham descent proof of the filtered calculation and a Rees–Fedosov reconstruction based on Lie-algebroid Hochschild-chain formality. In the second route the formal-to-classical comparison is made at every finite symbol order, while convergence for the actual continuous algebra remains the quoted Benameur–Nistor input. The Rees parameter is not confused with an additional geometric circle. We separate this result from the Hochschild homology of an unreduced operator algebra, which depends on the chosen residual ideal and completion.

# 1 The precise statement

The phrase *the algebra generated by b-pseudodifferential operators* is ambiguous. At least four choices affect Hochschild homology: the allowed operator orders, the allowed powers of a boundary defining function, the residual ideal, and the locally convex completion. We use the standard algebra for which the homology has a finite-dimensional geometric description. It is the b-groupoid specialization of the Laurent symbol algebras studied in [8, 2, 1].

For one boundary hypersurface, the standard b-groupoid has underlying set

$$\mathcal{G}_b = X^\circ \times X^\circ \ \sqcup \bigsqcup_{H \in \pi_0(\partial X)} H \times H \times \mathbb{R}.$$

On a boundary component, composition adds the last coordinate. Its smooth structure is fixed by the normal ratio $t = \log(x/x')$, and its Lie algebroid is $A(\mathcal{G}_b) = {}^bTX$. For corners, one uses the standard groupoid with an $\mathbb{R}^j$-factor over each open codimension-$j$ face [1].

Let $\mathcal{O}(X)$ be the ring generated by $C^\infty(X)$ and $x_H^{-1}$ for all boundary hypersurfaces $H$. Benameur and Nistor define

$$\mathcal{A}_{b,L}(X;E) := \mathcal{O}(X)\left(\Psi^{\infty}(\mathcal{G}_b;r^*E)/\Psi^{-\infty}(\mathcal{G}_b;r^*E)\right)$$

with their topologically filtered topology and completed Hochschild complex [2, 1]. The vector representation identifies this with the usual properly supported Laurent b-complete symbols. In particular, one may write its underlying filtered union heuristically as

$$\frac{\bigcup_{m,\ell} x^{-\ell}\Psi_b^m(X;E)}{\bigcup_\ell x^{-\ell}\Psi_b^{-\infty}(X;E)},$$

but this quotient notation does not by itself specify the topology used in the theorem.

Changing the defining function from $x$ to $x' = e^f x$ gives

$$(x')^{-\ell} = e^{-\ell f}x^{-\ell}.$$

Hence the change only multiplies a Laurent generator by a smooth invertible function, and the algebra, its filtration, and the answer below are canonically independent of this choice. For corners one allows an arbitrary finite Laurent monomial $\prod_H x_H^{-\ell_H}$, one factor for each boundary hypersurface.

Set $Y = {}^bS^*X = ({}^bT^*X \setminus 0)/\mathbb{R}_+$ and $Z = Y|_{\partial X}$. Thus $\partial Y = Z$. The symbol-order circle is denoted $S^1_\sigma$.

**Theorem 1.1** (Hochschild homology)**.** *For every $k \geq 0$,*

$$\boxed{\operatorname{gr} \mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X;E)) \cong H_L^{2n-k}(Y \times S^1_\sigma;\mathbb{C}).} \tag{1}$$

*After choosing splittings of the finite homology filtration, this gives a noncanonical vector-space isomorphism*

$$\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X;E)) \cong H_L^{2n-k}(Y \times S^1_\sigma;\mathbb{C}).$$

*The answer is independent of $E$. If $X$ has one embedded boundary hypersurface, then*

$$\begin{aligned} \mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}(X;E)) \cong{}& H^{2n-k}(Y) \oplus H^{2n-k-1}(Y) \\ &\oplus H^{2n-k-1}(Z) \oplus H^{2n-k-2}(Z), \end{aligned} \tag{2}$$

*where a cohomology group in negative degree is zero. In particular the homology vanishes for $k > 2n$.*

The associated-graded statement in (1) is intrinsic. The four-summand formula uses choices of Laurent face splitting and the Künneth splitting for $S^1_\sigma$.

$$\begin{tikzcd} & \mathcal{A}_{b,L} \arrow[r, "\text{order filtration}"] & \operatorname{gr} \mathcal{A}_{b,L} \arrow[r, "\text{HKR}"] & \Omega_L^*({}^bT^*X \setminus 0) \arrow[d, "\delta_{\pi_b}"] \\ \mathrm{HH}_*^{\mathrm{cont}}(\mathcal{A}_{b,L}) & & H_L^{2n-*}(Y \times S^1_\sigma) \arrow[ll, "E^2=E^\infty"'] & H_*^{\pi_b,L}({}^bT^*X \setminus 0) \arrow[l, "\text{symplectic duality}"'] \end{tikzcd}$$

Figure 1: The computation passes from the filtered noncommutative algebra to its commutative associated graded, then from Poisson homology to Laurent de Rham cohomology.

## 2 Continuous chains and the two filtrations

The topology is part of the theorem. For a topologically filtered algebra $A$, Benameur–Nistor do *not* take the completed projective tensor power of the whole locally convex space: multiplication need not be jointly continuous for that topology. In their notation the degree-$k$ chain space is

$$H_k(A) = \varinjlim_N \left(F^N_{N,N}A\right)^{\widehat{\otimes}_\pi (k+1)}. \tag{3}$$

Here $F^N_{N,N}A$ is the diagonal cofinal family of filtration pieces from [1, Definition 1]; equivalently, one first takes the projective completion in the decreasing symbol-order direction and only then enlarges the support and Laurent bounds by an inductive limit. We set

$$C_k^{\mathrm{cont}}(\mathcal{A}_{b,L}) := H_k(\mathcal{A}_{b,L}).$$

Whenever a tensor product below carries no decoration it denotes an elementary tensor in the dense filtered subcomplex of (3), not an ordinary completed tensor power of the whole algebra.

For a chain $a_0 \otimes \cdots \otimes a_k$, the Hochschild boundary is

$$\begin{aligned} b(a_0 \otimes \cdots \otimes a_k) = & \sum_{j=0}^{k-1} (-1)^j a_0 \otimes \cdots \otimes a_j a_{j+1} \otimes \cdots \otimes a_k \\ & + (-1)^k a_k a_0 \otimes a_1 \otimes \cdots \otimes a_{k-1}. \end{aligned}$$

Multiplication is continuous on fixed filtration pieces, so $b$ extends to the completed complex and satisfies $b^2 = 0$.

Laurent order and symbol order play different roles. The first is built into $\mathcal{O}(X)$; symbol order produces the spectral sequence. On every fixed piece in (3), filter elementary tensors by

$$\operatorname{ord}(a_0) + \cdots + \operatorname{ord}(a_k) \leq p,$$

take the projective closure there, and then pass through the same projective/inductive limits. Denote the resulting subspace by $\mathcal{F}_p C_k$. Then $b(\mathcal{F}_p C_k) \subseteq \mathcal{F}_p C_{k-1}$. Exhaustivity, completeness, and the convergence hypotheses are those verified in the Benameur–Nistor topologically filtered theorem. Borel quantization explains asymptotic completeness of symbols, but it is not by itself a convergence proof for an arbitrarily completed Hochschild complex.

*Discussion.* **Why divide by smoothing operators?** The principal-symbol filtration sees an operator only through its full asymptotic symbol. Smoothing kernels form the kernel of that passage. Keeping them gives an extension problem, treated separately in Section 12; it is not a harmless change of notation.

# 3 The b-cotangent geometry

The Lie algebra of b-vector fields is

$$\mathcal{V}_b(X) = \{V \in C^\infty(X; TX) : V \text{ is tangent to } \partial X\}.$$

It is the space of smooth sections of a vector bundle ${}^bTX \to X$. Near a boundary point, choose product coordinates

$$(x, y_1, \ldots, y_{n-1}).$$

Then

$$\mathcal{V}_b(X) = C^\infty(X)\text{-span}\{x\partial_x, \partial_{y_1}, \ldots, \partial_{y_{n-1}}\}.$$

The dual local frame of ${}^bT^*X$ is

$$\frac{dx}{x}, \quad dy_1, \ldots, dy_{n-1}.$$

Thus a b-covector has the form

$$\xi \frac{dx}{x} + \eta \cdot dy.$$

On ${}^bT^*X$ the tautological b-one-form and its differential are

$$\alpha_b = \xi \frac{dx}{x} + \eta \cdot dy, \qquad \omega_b = d\alpha_b = d\xi \wedge \frac{dx}{x} + d\eta \wedge dy.$$

The form $\omega_b$ is nondegenerate as a b-two-form. Its inverse is the b-Poisson tensor

$$\pi_b = x\partial_x \wedge \partial_\xi + \sum_j \partial_{y_j} \wedge \partial_{\eta_j}.$$

This is the geometric bracket appearing in the first nonzero differential of the Hochschild spectral sequence.

**Proposition 3.1** (Anchor bridge)**.** *The anchor $a : {}^bTX \to TX$ becomes an isomorphism after adjoining Laurent coefficients:*

$$\mathcal{O}(X) \otimes_{C^\infty(X)} \Gamma({}^bTX) \xrightarrow{a} \mathcal{O}(X) \otimes_{C^\infty(X)} \Gamma(TX).$$

*After dualizing, this is not a smooth vector-bundle isomorphism at the boundary. It is an isomorphism only after Laurent localization:*

$$\mathcal{O}(X) \otimes C^\infty({}^bT^*X) \cong \mathcal{O}(X) \otimes C^\infty(T^*X)$$

*as fiberwise-polynomial Poisson algebras, extended by completion to the homogeneous Laurent symbol complexes. Consequently the anchor lemmas of [1] give a chain-level isomorphism of the corresponding homogeneous Laurent de Rham complexes and hence*

$$H^*_L({}^bS^*X \times S^1_\sigma) \cong H^*_L(S^*X \times S^1_\sigma).$$

*Proof.* Locally the anchor sends $x\partial_x$ to the ordinary vector field $x\partial_x$ and fixes $\partial_{y_j}$. Its localized inverse uses the Laurent coefficient $x^{-1}$: $\partial_x \mapsto x^{-1}(x\partial_x)$. The map preserves brackets, so its localized dual preserves the linear Poisson brackets. Taking the degree-zero homogeneous subcomplex and adjoining the order-circle generator therefore gives the stated cohomology isomorphism. This is the one-fibre specialization of the anchor lemmas in [1]; no isomorphism ${}^bT^*X \cong T^*X$ of smooth bundles is being asserted. □

# 4 The associated graded algebra

Filter $\mathcal{A}_{b,L}$ by operator order:

$$F_m\mathcal{A}_{b,L} = \text{image}\left(\bigcup_{\ell\geq 0} x^{-\ell}\Psi^m_b(X;E)\right), \qquad m \in \mathbb{Z}.$$

The filtration is exhaustive, separated in the complete-symbol sense, and multiplicative. Principal symbols give

$$\mathrm{gr}_m\,\mathcal{A}_{b,L} \cong \mathcal{O}_L({}^bT^*X \setminus 0; \mathrm{End}\,E)_m,$$

the Laurent functions that are homogeneous of degree $m$ in the fibers. In a b-Darboux chart, a choice of local Weyl quantization gives the expansion

$$a\#c \sim ac + \frac{1}{2i}\{a,c\}_b + \text{terms two orders lower}, \tag{4}$$

where changing quantization changes the individual lower-order symmetric terms. Without a chosen global symplectic connection, the full Weyl formula is not coordinate invariant. What is invariant, and all that is used below, is the commutator symbol

$$\sigma_{m+m'-1}([A,C]) = i^{-1}\{a,c\}_b.$$

This one-order drop is the origin of the first nonzero differential.

**Lemma 4.1** (Morita reduction)**.** *The coefficient bundle $E$ does not change the Hochschild homology.*

*Proof.* The global module of complete symbols with coefficients in $E$ is a finitely generated projective equivalence bimodule between the scalar algebra and its End $E$-coefficient algebra. The associated matrix-trace and corner maps are continuous and filtration preserving, and the standard matrix-unit homotopy proves Morita invariance. This is a global Morita equivalence; no partition of unity is used to glue local chain homotopies. The topologically filtered version, including graded bundles, is the coefficient-independence lemma of [2, 1]. □

We therefore suppress $E$. The Hochschild–Kostant–Rosenberg map on the associated graded algebra is

$$\chi(a_0 \otimes \cdots \otimes a_j) = \frac{1}{j!} a_0\, da_1 \wedge \cdots \wedge da_j.$$

**Proposition 4.2** (The $E^1$-page)**.** *The HKR map identifies the first page of the order spectral sequence with Laurent differential forms on ${}^bT^*X \setminus 0$, decomposed by fiber homogeneity.*

*Proof.* On each b-cotangent chart the associated graded product is ordinary multiplication of scalar homogeneous symbols. Antisymmetrization gives the inverse to $\chi$ on Hochschild homology:

$$a_0, da_1 \wedge \cdots \wedge da_j \longmapsto \sum_{\tau \in S_j} \operatorname{sgn}(\tau) a_0 \otimes a_{\tau(1)} \otimes \cdots \otimes a_{\tau(j)}.$$

The displayed antisymmetrization is not a literal chain-level inverse; the two composites become inverse after the Koszul homotopy. For the topologically filtered Laurent groupoid algebra, continuity, completion, and globalization are the HKR proposition used by Benameur–Nistor [2, 1]. It may be proved through the sheaf of local Koszul resolutions and its fine resolution, but a bare partition of unity does not patch chain homotopies. Fiber dilation commutes with the published quasi-isomorphism, so it retains homogeneity and gives

$$E^1_{p,q} \cong \Omega_L^{p+q}({}^bT^*X \setminus 0)_p,$$

with the conventional reindexing determined by total symbol order. □

# 5 The first differential is Poisson homology

For classical b-symbols $a$ and $c$, symbolic composition gives

$$\sigma_{m+m'-1}([A, C]) = \frac{1}{i}\{a, c\}_b.$$

Consequently the first differential after HKR is, up to the harmless factor $i$, the Poisson differential

$$\delta_b = [\iota_{\pi_b}, d] = \iota_{\pi_b} d - d\iota_{\pi_b}.$$

Explicitly,

$$\begin{aligned}\delta_b(a_0, da_1 \wedge \cdots \wedge da_j) = &\sum_{r=1}^{j} (-1)^{r+1} \{a_0, a_r\}_b\, da_1 \wedge \cdots \widehat{da_r} \cdots \wedge da_j \\ &+ \sum_{1 \le r < s \le j} (-1)^{r+s} a_0\, d\{a_r, a_s\}_b \wedge da_1 \wedge \cdots \widehat{da_r} \cdots \widehat{da_s} \cdots \wedge da_j. \qquad (5)\end{aligned}$$

To see this differential directly, insert the order-one-lower part of (4) into each multiplication in the Hochschild boundary. The symmetric terms cancel after HKR antisymmetrization, while the commutator terms combine to (5).

Let $N = 2n$ be the dimension of ${}^bT^*X$, put $\mu_b = \omega_b^n/n!$, and let $\pi_b^\sharp : \Omega^j \to \Lambda^j T$ be contraction of every covector with $\pi_b$. Symplectic duality is the map

$$*_b : \Omega_L^j({}^bT^*X \setminus 0) \longrightarrow \Omega_L^{N-j}({}^bT^*X \setminus 0), \qquad *_b\alpha = \iota_{\pi_b^\sharp \alpha}\mu_b,$$

and $*_b^2 = 1$ up to the standard degree sign. Applying the two sides to a monomial in the Darboux b-frame

$$\frac{dx}{x}, dy_1, \ldots, dy_{n-1}, d\xi, d\eta_1, \ldots, d\eta_{n-1}$$

gives, with the displayed convention for $\delta_b$,

$$*_b\delta_b = (-1)^{j+1} d *_b \quad \text{on } \Omega_L^j. \tag{6}$$

The calculation remains valid at $x = 0$: the displayed frame is smooth as a b-frame, and both operators preserve finite Laurent pole order. Therefore a $\delta_b$-cycle of degree $j$ is carried to a closed Laurent form of degree $2n - j$, while boundaries are carried to exact forms.

**Corollary 5.1.** *The Poisson homology on the $E^2$-page is Laurent de Rham cohomology with complementary degree:*

$$H_j^{\pi_b, L}({}^bT^*X \setminus 0) \cong H_L^{2n-j}({}^bT^*X \setminus 0).$$

## 6 Homogeneity and the symbol-order circle

Choose a positive fiber norm $r$ on ${}^bT^*X \setminus 0$. Then

$${}^bT^*X \setminus 0 \cong Y \times \mathbb{R}_+, \qquad (q, r) \longmapsto rq.$$

The order index in the spectral sequence and the degree reversal under $*_b$ combine so that its relevant de Rham complex is the total homogeneity-zero complex. This bookkeeping matters: $\delta_b$ lowers fiber homogeneity by one, whereas $*_b$ changes both form degree and homogeneity. After the complementary-degree regrading, the surviving weight is zero, as in the complete-symbol theorem of Brylinski–Getzler and Benameur–Nistor.

The zero-weight reduction itself is elementary. If $\mathcal{E} = r\partial_r$ is the Euler vector field and a form $\alpha$ has nonzero homogeneity $w$, Cartan's formula gives

$$(d\iota_{\mathcal{E}} + \iota_{\mathcal{E}} d)\alpha = \mathcal{L}_{\mathcal{E}}\alpha = w\alpha.$$

Thus $w^{-1}\iota_{\mathcal{E}}$ contracts every nonzero weight.

At weight zero, restriction to $r = 1$ gives a unique splitting

$$\alpha(r, q) = \alpha_0(q) + \frac{dr}{r} \wedge \alpha_1(q).$$

Since $d(dr/r) = 0$, its differential is

$$d\alpha = d_Y\alpha_0 - \frac{dr}{r} \wedge d_Y\alpha_1.$$

Thus the weight-zero complex is the direct sum of the de Rham complex of $Y$ and a copy shifted by one degree. The complete-symbol order is an integer grading with no preferred logarithm; cohomologically this two-term complex is canonically written as the de Rham complex of an auxiliary circle:

$$\alpha_0 + \frac{dr}{r} \wedge \alpha_1 \longleftrightarrow \alpha_0 + d\theta_\sigma \wedge \alpha_1.$$

The formal order variable $dr/r$ is represented cohomologically by the generator $d\theta_\sigma$ of a circle. Hence

$$H^j_L({}^bT^*X \setminus 0)_{\mathrm{hom}=0} \cong H^j_L(Y \times S^1_\sigma). \tag{7}$$

This is the source of the extra $S^1$ in the answer; it is not a geometric circle in the fibers of ${}^bT^*X$, and no choice of fiber norm affects its cohomology class.

## 7 Laurent cohomology and the boundary face

Let $M$ be a compact manifold with one embedded boundary hypersurface and defining function $x$. The Laurent de Rham complex is

$$\Omega^j_L(M) = \bigcup_{\ell \geq 0} x^{-\ell}\Omega^j(M), \qquad d : \Omega^j_L(M) \to \Omega^{j+1}_L(M).$$

**Proposition 7.1** (Face decomposition)**.** *There is an isomorphism*

$$H^j_L(M) \cong H^j(M) \oplus H^{j-1}(\partial M).$$

*The boundary summand is represented by $\widetilde{\beta} \wedge d(\chi \log x)$, where $\beta$ is a closed $(j-1)$-form on $\partial M$, $\widetilde{\beta}$ is a collar extension, and $\chi$ is a cutoff supported in the collar and equal to one near the boundary.*

*Proof.* Work in a collar and write a Laurent form uniquely as

$$\alpha(x) + \frac{dx}{x} \wedge \beta(x),$$

where the tangential forms $\alpha(x), \beta(x)$ are Laurent families with a finite principal part and a smooth Taylor tail. Filter the collar complex by pole order and then by normal jet. On its associated graded, a term has a definite normal weight $q$. For $q \neq 0$, Cartan's identity gives the contracting homotopy

$$h_q = q^{-1}\iota_{x\partial_x}, \qquad dh_q + h_q d = 1.$$

Induction over the finite negative weights removes the principal part; the remaining positive Taylor tail is smooth and belongs to the ordinary de Rham complex. This jet-filtration argument, rather than freezing all coefficients at $x = 0$, is the Laurent Poincaré lemma. The only new normal class is the weight-zero generator $dx/x$.

In particular, after writing

$$\alpha = \alpha_0 + \frac{dx}{x} \wedge \beta_0 + \text{terms of nonzero normal weight},$$

the residue is

$$\mathrm{Res}_{\partial M} : \Omega^j_L(M) \longrightarrow \Omega^{j-1}(\partial M), \qquad \mathrm{Res}_{\partial M}(\alpha) = \beta_0|_{\partial M}.$$

A direct calculation in the displayed order gives

$$\operatorname{Res}(d\alpha) = -d_{\partial M} \operatorname{Res}(\alpha).$$

Thus residue is a chain map to the degree-shifted boundary complex. The Laurent Poincaré lemma identifies its homotopy kernel with $\Omega^*(M)$, yielding the quasi-isomorphism

$$\Omega_L^*(M) \simeq \Omega^*(M) \oplus \Omega^{*-1}(\partial M).$$

If $\beta$ is closed of degree $j-1$, let $p : [0,\varepsilon)_x \times \partial M \to \partial M$ be the collar projection. Choose $\chi$ with support in this collar and choose $\widetilde{\beta} = p^*\beta$ on an open set containing $\operatorname{supp} d(\chi \log x)$. Extend it arbitrarily beyond that set. Then $d\widetilde{\beta} = 0$ everywhere that $d(\chi \log x)$ is nonzero, so the representative below is closed. This collar construction gives the cohomological splitting

$$\beta \longmapsto (-1)^{j-1}\widetilde{\beta} \wedge d(\chi \log x).$$

Its residue is $\beta$. If $x' = e^f x$, then $d\log x' = d\log x + df$; changing the defining function, collar, or cutoff changes this splitting by an ordinary/exact correction. Hence the direct sum is natural as a filtered extension but its explicit vector-space splitting is not canonical. □

*Discussion.* **Where do higher poles go?** A pole such as $x^{-q-1}dx$, $q > 0$, looks more singular than $dx/x$, but it is $-q^{-1}d(x^{-q})$. Higher Laurent coefficients occur in the complex and are essential for closure under composition; they add no new normal cohomology. Only the logarithmic coefficient survives.

Apply Proposition 7.1 to $M = Y \times S^1_\sigma$, whose boundary is $Z \times S^1_\sigma$. Then use

$$H^r(W \times S^1) \cong H^r(W) \oplus H^{r-1}(W).$$

For $r = 2n - k$ this gives exactly the four summands in (2).

$$\begin{array}{c}
H^r_L(Y \times S^1_\sigma) \\
\Big\| \\
H^r(Y \times S^1_\sigma) \oplus H^{r-1}(Z \times S^1_\sigma) \\
\Big\| \\
\big(H^r(Y) \oplus H^{r-1}(Y)\big) \oplus \big(H^{r-1}(Z) \oplus H^{r-2}(Z)\big)
\end{array} \qquad r = 2n - k$$

Figure 2: The four summands come from the ordinary class, the symbol-order logarithm $d\log r$, the boundary logarithm $d\log x$, and their product.

For a manifold with corners, the same local argument has one generator $d\log x_H$ for every boundary hypersurface $H$ through a point. If $\mathcal{F}_j(M)$ denotes the codimension-$j$ faces, then

$$H^r_L(M) \cong \bigoplus_{j\geq 0} \bigoplus_{F\in\mathcal{F}_j(M)} H^{r-j}(F; o(N_F)). \tag{8}$$

Here $o(N_F)$ is the orientation line of the conormal directions to $F$. If the hypersurfaces have been globally cooriented and ordered, these lines are trivialized and the untwisted formula results.

Indeed, apply the one-face residue successively. Residues for distinct hypersurfaces anticommute with the graded sign, so the summand attached to $F = H_1 \cap \cdots \cap H_j$ is represented by

$$\widetilde{\gamma} \wedge d\log x_{H_1} \wedge \cdots \wedge d\log x_{H_j}, \qquad [\gamma] \in H^{r-j}(F).$$

The product of the nonzero-weight contractions gives an acyclic complement. Changing the order of the hypersurfaces changes the orientation sign of the exterior logarithmic factors; the local system $o(N_F)$ records exactly this ambiguity. Iterated collar extensions provide a noncanonical splitting. This proves (8) rather than merely suggesting it from the one-boundary case.

# 8 Collapse and assembly of the computation

Combining HKR, the Poisson differential, symplectic duality, and homogeneity gives

$$E^2_k \cong H^{2n-k}_L(Y \times S^1_\sigma).$$

The remaining input is the collapse theorem for Laurent complete symbols.

**Theorem 8.1** (Benameur–Nistor collapse theorem; quoted)**.** *Let $G$ satisfy the anchor and local-triviality hypotheses of [1, equations (13)–(14)]. For its Benameur–Nistor topologically filtered Laurent complete-symbol algebra, the order spectral sequence converges and degenerates at $E^2$. These hypotheses hold for the canonical b-groupoid $\mathcal{G}_b$ used in this manuscript.*

This is the global input from [2, 1]. It is stronger than a degree-counting assertion: higher differentials could a priori connect different symbol orders while preserving total degree. The theorem rules them out for the completed classical-symbol algebra.

The published collapse proof first treats a locally trivial product family, where the Künneth theorem produces Hochschild classes realizing every $E^2$-class. Naturality then transfers survival to the completed product algebra, and multiplication by compactly supported base functions localizes the argument to trivializing sets. This proves vanishing of all higher differentials and the convergence bounds. It is not a "symplectic homotopy kills higher differentials" argument; symplectic duality computes $E^2$, whereas locality and Künneth prove collapse.

**Proposition 8.2** (Source dictionary)**.** *The algebra used here is exactly the one-face b-groupoid specialization of the Benameur–Nistor theorem.*

*Proof.* The dictionary has six entries.

*Groupoid.* The differential groupoid is $\mathcal{G}_b$, with unit space $X$ and Lie algebroid ${}^bTX$.

*Laurent multipliers.* The coefficient ring is precisely $\mathcal{O}(X) = C^\infty(X)[x_H^{-1}]$.

*Algebra and topology.* The algebra is $\mathcal{O}(X)(\Psi^\infty(\mathcal{G}_b; E)/\Psi^{-\infty}(\mathcal{G}_b; E))$, equipped with the topologically filtered completion of [1]; its continuous Hochschild complex is the one used in their theorem.

*Anchor assumption.* With base reduced to a point, the fibre condition is automatic. The remaining assumption is exactly the Laurent anchor isomorphism proved in the anchor-bridge proposition.

*Local triviality and coefficients.* Benameur–Nistor prove that the anchor assumption supplies their required local topologically filtered models. Their global Morita lemma removes the coefficient bundle $E$.

*Spectral sequence.* Their filtered HKR proposition identifies $E^1$; their homogeneous Laurent-Poisson theorem gives $E^2$; and their collapse/convergence lemma identifies $E^\infty$ with continuous Hochschild homology. Thus no independently asserted LF, patching, or Borel splitting hypothesis remains to be checked. □

*Proof of Theorem 1.1.* Proposition 8.2 gives a convergent spectral sequence. The principal-symbol identification and HKR theorem give

$$E^1 \cong \Omega_L^*({}^bT^*X \setminus 0)_{\mathrm{hom}},$$

and expansion (4) identifies $d^1$ with the Poisson boundary $\delta_b$. Symplectic duality (6) reverses degree, and the Euler contraction removes every nonzero total weight. Therefore

$$E_k^2 \cong H_L^{2n-k}(Y \times S_\sigma^1).$$

The collapse theorem gives $E^2 = E^\infty$, so this is the associated graded of $\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L})$.

A Borel quantization splits the symbol filtration on the algebra, but it does not by itself split the induced filtration on homology. The Benameur–Nistor theorem supplies the convergent filtration; over $\mathbb{C}$, choosing linear complements of its finitely many nonzero graded pieces gives the stated noncanonical vector-space isomorphism. The canonical assertion is the associated-graded identification (1). Finally, Proposition 7.1 and Künneth for $S_\sigma^1$ give the four summands in (2). □

*Discussion.* **Why is convergence not automatic?** For a merely algebraic union of symbol orders, an infinite correction can escape the complex and an $E^\infty$-class need not lift to a chain. Borel summation and the strict Laurent inductive limit are ingredients in the topologically filtered axioms: the former realizes asymptotic symbols and the latter keeps every individual chain at finite pole order. They do not by themselves prove convergence. Convergence to Hochschild homology is the quoted Benameur–Nistor spectral-sequence theorem, including its uniform vanishing-range verification.

## 9 A Čech–de Rham reconstruction

This section gives a second organization of the proof. It replaces the global HKR assertion by a local calculation followed by descent. It still uses the Benameur–Nistor convergence-and-collapse theorem; Čech descent alone computes the $E^2$-page, not the abutment.

Choose a finite good cover $\mathcal{U} = \{U_i\}$ of $Y = {}^bS^*X$. Near $Z = \partial Y$, take product sets $[0, \varepsilon)_x \times V_i$, and require every nonempty intersection to be contractible and collar adapted. Write $\widetilde{U}_I = U_I \times \mathbb{R}_+$ for the corresponding conic open set in ${}^bT^*X \setminus 0$.

Let $\mathcal{C}_q$ be the sheaf whose sections over $U$ are the Benameur–Nistor completed degree-$q$ chains of Laurent complete symbols microlocally supported in $\widetilde{U}$. Thus one sheafifies on each diagonal piece $F_{N,N}^N$ first, takes its completed projective tensor power, and only then takes the strict inductive limit in $N$. This order of operations is the sheaf version of (3).

Form the double complex

$$K^{p,q} = \check{C}^p(\mathcal{U}, \mathcal{C}_q), \qquad D = \delta_{\check{C}} + (-1)^p b. \tag{9}$$

The total degree is $q - p$, since $b$ lowers Hochschild degree and the Čech differential raises $p$.

**Lemma 9.1** (Strict descent for completed chains)**.** *The augmentation*

$$C_\bullet^{\mathrm{cont}}(\mathcal{A}_{b,L}) \longrightarrow \operatorname{Tot} K^{\bullet,\bullet}$$

*is a filtration-preserving quasi-isomorphism. It remains a quasi-isomorphism after taking any fixed order quotient.*

*Proof.* Fix a diagonal filtration piece and a Hochschild degree. Local complete symbols of bounded upper order are nuclear Fréchet spaces, and the same is true of their completed projective tensor powers. A smooth partition of unity $\{\rho_i\}$ subordinate to $\mathcal{U}$, lifted homogeneously to the conic sets, gives the usual contracting homotopy of the augmented Čech row: multiply the first tensor factor by $\rho_i$, restrict, and sum in the new index. The homotopy is continuous and does not increase total symbol order or Laurent pole order.

This row homotopy need not commute with the Hochschild boundary; that is not required. It proves strict exactness of every augmented Čech row. Because the homotopy is defined uniformly on the diagonal pieces, it passes through their projective completions and strict inductive limit. The spectral sequence obtained by taking Čech cohomology first is therefore concentrated in degree zero and proves the assertion. The same argument works after quotienting by any lower-order piece. □

Filter (9) by total symbol order. On the associated graded, multiplication is commutative. The local HKR map and the local Koszul homotopy identify the first page with

$$\check{C}^p\big(\mathcal{U}, \Omega_L^q({}^bT^*X \setminus 0)_{\mathrm{hom}}\big). \tag{10}$$

These maps commute with restriction. The first order-lowering part of the symbol product gives $i^{-1}\delta_b$, exactly as in (5). Hence the next differential on (10) is the Poisson differential.

For a closed manifold these are exactly the local filtered steps in the Brylinski–Getzler calculation [3]: complete symbols pass to homogeneous functions, HKR gives forms, and the first quantum correction is the Poisson boundary. The present descent proof adds the Laurent normal-weight contraction, the residue complex at $Z$, and the strict topologically filtered completion required at the boundary.

**Proposition 9.2** (Čech computation of the second page)**.** *The total filtered complex* (9) *has*

$$E_k^2 \cong H_L^{2n-k}(Y \times S^1_\sigma; \mathbb{C}).$$

*This identification is independent of the good cover and the partition of unity.*

*Proof.* Symplectic duality (6) converts the Poisson differential in (10) to the Laurent de Rham differential with complementary degree. The sheaves of Laurent forms with a fixed finite pole bound are fine. Their union is still Čech acyclic for the finite cover, because multiplication by the partition of unity preserves every pole bound. Thus the Čech direction contributes no positive-degree cohomology and the hypercohomology is the cohomology of global Laurent forms.

The Euler homotopy removes nonzero fiber weights. At weight zero a form is $\alpha_0 + (dr/r) \wedge \alpha_1$, so (7) identifies the remaining complex with Laurent forms on $Y \times S^1_\sigma$. Refining the cover gives a common refinement, and the two refinement maps are homotopic on the augmented Čech resolution. This proves independence of all auxiliary choices. □

The boundary term can also be read directly in the descent complex. One must not write the generally false short exact sequence $0 \to \Omega^*(Y) \to \Omega_L^*(Y) \to \Omega^{*-1}(Z) \to 0$: a Laurent form with zero residue can still have higher poles. Instead let $\mathcal{K}^* = \ker(\mathrm{Res}_Z)$. The nonzero normal-weight contraction in the proof of Proposition 7.1 gives a quasi-isomorphism

$$\Omega^*(Y) \longrightarrow \mathcal{K}^*.$$

The exact sequence

$$0 \longrightarrow \mathcal{K}^* \longrightarrow \Omega_L^*(Y) \xrightarrow{\mathrm{Res}_Z} \big(\Omega^{*-1}(Z), -d_Z\big) \longrightarrow 0$$

therefore yields the derived decomposition

$$R\Gamma(Y;\Omega_L^*) \simeq R\Gamma(Y;\Omega^*) \oplus R\Gamma(Z;\Omega^{*-1}), \tag{11}$$

after choosing a collar splitting. This is precisely the interior/boundary decomposition used in (2); it also shows why the splitting, unlike the residue filtration, is not canonical.

Finally apply the quoted theorem in Section 8 to the order filtration on (9). Lemma 9.1 identifies its abutment with $\mathrm{HH}_*^{\mathrm{cont}}(\mathcal{A}_{b,L})$, while the proposition identifies its second page. Benameur–Nistor degeneration gives $E^2 = E^\infty$, completing the Čech–de Rham proof of Theorem 1.1. Thus descent replaces the globalization part of the original proof, but it does not manufacture convergence or kill higher differentials by itself.

# 10 A Rees–Fedosov reconstruction

The third route makes the deformation behind symbol composition explicit. It has two logically separate parts: a formal Lie-algebroid formality theorem and a finite-order Rees comparison with classical symbols. Passage to the completed continuous complex remains the convergence input quoted in Section 8.

This is also the deformation-theoretic refinement of the Brylinski–Getzler filtration: their principal-symbol and Poisson differentials are the first two coefficients of the Rees star product below [3].

## 10.1 The Rees deformation

The algebraic Rees algebra of the order filtration is

$$\mathcal{R}_F(\mathcal{A}_{b,L}) = \bigoplus_{m\in\mathbb{Z}} t^m F_m \mathcal{A}_{b,L} \subset \mathcal{A}_{b,L}[t,t^{-1}].$$

It records both specialization to the associated graded and the return to complete symbols at $t = 1$. Since $[F_m, F_{m'}] \subset F_{m+m'-1}$, its commutator is divisible by the Rees parameter in an associated-graded chart. Equivalently, for a symbol of fixed upper order $a \sim a_m + a_{m-1} + a_{m-2} + \cdots$, factor off the bookkeeping degree $t^m$ and encode the descending expansion by

$$a_m + \hbar a_{m-1} + \hbar^2 a_{m-2} + \cdots . \tag{12}$$

The power $t^m$ records the absolute order. After it is factored off in a fixed sector, we write the same Rees parameter as $\hbar$; its successive powers record loss of order. Thus (12) is an $\hbar$-adic chart on one upper-order sector, not a global $\mathbb{C}[[\hbar]]$-module structure on the union of all orders.

On Laurent functions on $M = {}^bT^*X \setminus 0$, full symbol composition in this chart becomes

$$a \star c = \sum_{j\geq 0} \hbar^j C_j(a,c), \qquad C_0(a,c) = ac, \tag{13}$$

where $C_j$ is b-bidifferential and lowers the unscaled symbol order by $j$. Its antisymmetric first term is

$$C_1(a,c) - C_1(c,a) = i^{-1}\{a,c\}_b,$$

so $[a,c]_\star = (\hbar/i)\{a,c\}_b + O(\hbar^2)$.

The space $M$ has the symplectic Lie algebroid ${}^bTM$ with symplectic form $\omega_b$. Choose a torsion-free b-symplectic connection and use the homogeneous cotangent Fedosov construction, so that positive fiber dilation acts equivariantly. The Fedosov Weyl bundle has a flat differential of the form

$$D_F = -\delta + \nabla + \frac{i}{\hbar}\,\mathrm{ad}(r_F), \qquad D_F^2 = 0.$$

Flat sections, transported to Laurent functions by the Fedosov–Taylor map, give a star product equivalent to (13). Symplectic Lie-algebroid Fedosov quantization, including logarithmic symplectic structures and corners, is developed by Nest–Tsygan [9]. Equivalent star products induce isomorphic Hochschild complexes, so the choice of connection does not affect the answer.

### 10.2 Hochschild-chain formality

Calaque–Dolgushev–Halbout prove Tsygan formality for Hochschild chains in the Lie-algebroid setting by combining Fedosov resolutions with the Kontsevich–Shoikhet local maps [4]; Dolgushev's smooth-manifold theorem is the corresponding untwisted prototype [5]. Twisting their module quasi-isomorphism by the Maurer–Cartan element defining (13) gives

$$(C_\bullet(\mathcal{O}_L(M)((\hbar)),\star), b_\star) \simeq (\Omega_L^\bullet(M)((\hbar)), \mathcal{L}_{\pi_\hbar}), \tag{14}$$

where $\pi_\hbar = \hbar\pi_b + O(\hbar^2)$ is the formal Poisson tensor and $\mathcal{L}_{\pi_\hbar} = [\iota_{\pi_\hbar}, d]$.

The formal tensor $\pi_\hbar$ is symplectic over $\mathbb{C}((\hbar))$; let $\omega_\hbar$ be its inverse. The formal symplectic star operator $*_{\omega_\hbar}$ satisfies the same Darboux calculation as (6) and converts $\mathcal{L}_{\pi_\hbar}$ to $d$, up to an invertible scalar and the degree sign. Therefore

$$\mathrm{HH}_k^{\mathbb{C}((\hbar))}(\mathcal{O}_L(M)((\hbar)),\star) \cong H_L^{2n-k}(M;\mathbb{C})((\hbar)). \tag{15}$$

The formality maps are b-polydifferential. Hence they preserve finite Laurent pole order: locally $x\partial_x(x^{-\ell}) = -\ell x^{-\ell}$, and tangential b-derivatives do not create a larger pole.

Equation (15) concerns the extended formal star algebra, before restriction to classical homogeneous expansions and before Borel realization. It must not be read as the complete-symbol answer. In particular, $H_L^*(M)$ itself has no order-circle factor: that factor appears only after the homogeneous order complex is formed.

*Remark* 10.1 (Where the order circle does, and does not, come from). The complex in (14) is relative to the base field $\mathbb{C}((\hbar))$. Thus $d\hbar = 0$, and it is incorrect to insert an additional class $d\hbar/\hbar$ into (15). The single class giving $S_\sigma^1$ is the radial zero-weight class $dr/r$ in Section 5.

One can instead compute absolute Hochschild homology over $\mathbb{C}$ for the Laurent extension of a Rees algebra. Then a class $d\log t$ can appear. The Rees grading ties the exponent of $t$ to fiber homogeneity, so on the diagonal order complex $d\log t$ and $-d\log r$ describe the same order direction. After specialization at $t=1$, only the radial class remains. Either convention yields one order circle, never two.

### 10.3 From formal series to classical complete symbols

Formal formality does not by itself compute the continuous homology of $\mathcal{A}_{b,L}$. What follows directly from the classical symbol calculus is a comparison at every finite depth of the order filtration.

**Proposition 10.2** (Finite-order Rees comparison)**.** *After fixing a homogeneous classical symbol map, for every $m \in \mathbb{Z}$ and $N \geq 0$, the first $N+1$ symbol components give a continuous isomorphism*

$$F_m\mathcal{A}_{b,L}/F_{m-N-1}\mathcal{A}_{b,L} \cong \bigoplus_{j=0}^{N} \mathcal{O}_L(M)_{m-j}. \tag{16}$$

*Under* (16), *operator composition is the star product* (13) *modulo* $\hbar^{N+1}$. *These isomorphisms are compatible with truncation, completed projective tensor powers on each diagonal piece, and the Hochschild boundary. Consequently the Rees and classical order spectral sequences have the same* $E^1$*-page and the same Poisson differential* $d^1$.

*Proof.* There are three points to check.

*Algebraic realization.* Taylor expansion in symbol order gives (16); no infinite summation is needed at finite $N$. The b-symbol composition formula shows coefficient by coefficient that the two products agree modulo $\hbar^{N+1}$. Its first antisymmetric correction is the b-Poisson bracket, proving the assertion about $d^1$.

*Topology.* For a fixed Laurent bound, both sides of (16) are finite direct sums of nuclear Fréchet spaces. Truncation and the coefficient maps are continuous. Taking completed projective tensor powers first and the strict inductive limit over diagonal pieces afterward is exactly the order of operations in (3); hence the comparison extends to the finite quotients of the continuous Hochschild complex.

*Inverse limit.* The b-symbol Borel lemma realizes every compatible formal homogeneous sequence by a classical polyhomogeneous symbol, uniquely modulo smoothing. It therefore identifies the projective limit of the algebraic truncations with complete symbols. This asymptotic completeness does not, by itself, prove that Hochschild homology commutes with the projective/inductive limits; that separate convergence assertion is exactly the topologically filtered theorem quoted in Section 8. □

Formality now explains the local differential beyond its first Poisson term, while Proposition 10.2 identifies the finite classical jets to which that calculation applies. On the homogeneous Poisson complex, the Euler contraction makes every nonzero weight acyclic. The zero-weight splitting is

$$\Omega^j_L(Y) \oplus \frac{dr}{r} \wedge \Omega^{j-1}_L(Y),$$

whose cohomology is $H^j_L(Y \times S^1_\sigma)$. We have proved:

**Theorem 10.3** (Rees–Fedosov reconstruction)**.** *The Rees deformation and Lie-algebroid Hochschild-chain formality compute the second page of the classical order spectral sequence as*

$$E^2_k \cong H^{2n-k}_L(Y \times S^1_\sigma; \mathbb{C}),$$

*and the Benameur–Nistor convergence-and-collapse theorem then gives the associated-graded and noncanonical vector-space statements of Theorem 1.1.*

The formal chain map organizes all local higher corrections, but it does not license specialization of formal Hochschild homology to continuous Hochschild homology. The finite-order comparison is elementary symbol calculus; convergence and collapse for the completed classical algebra are still the published global input. This is the precise scope of the second route.

# 11 Checks and consequences

## 11.1 The closed-manifold case

If $\partial X = \varnothing$, Laurent cohomology is ordinary cohomology and $Z = \varnothing$. Formula (2) reduces to

$$\mathrm{HH}^{\mathrm{cont}}_k(\mathcal{A}(X;E)) \cong H^{2n-k}(S^*X) \oplus H^{2n-k-1}(S^*X),$$

equivalently $H^{2n-k}(S^*X \times S^1_\sigma)$. This is the Brylinski–Getzler computation [3].

$$
\begin{tikzcd}
C_\bullet^{\mathrm{cont}}(\mathcal{A}_{b,L}) \arrow[r, "\text{Čech descent}"] \arrow[d, dashed, "\text{finite Rees jets}"'] & \operatorname{Tot}\check{C}^\bullet(\mathcal{U}, \mathcal{C}_\bullet) \arrow[r, "\mathrm{HKR}+\delta_b"] & \Omega_L^{2n-\bullet}(M)_0 \\
C_\bullet(\mathcal{O}_L(M)((\hbar)), \star) \arrow[r, "\text{Fedosov formality}"'] & (\Omega_L^\bullet(M)((\hbar)), \mathcal{L}_{\pi_\hbar}) \arrow[ur, "*_{\omega_\hbar}"'] &
\end{tikzcd}
$$

Figure 3: The Čech route globalizes the filtered HKR–Poisson calculation and then uses the published collapse theorem. The Rees route organizes the local deformation by chain-level formality; finite Rees jets compare it with classical symbols, and the same published theorem controls the completed limit.

### 11.2 Degree zero and residue traces

For the completed chain complex (3), degree-zero Hochschild homology is

$$\mathrm{HH}_0^{\mathrm{cont}}(\mathcal{A}) = \mathcal{A}/\operatorname{im}(b : C_1^{\mathrm{cont}} \to \mathcal{A}).$$

We abbreviate the image by $[\mathcal{A}, \mathcal{A}]_{\mathrm{cont}}$; it need not be closed. Its Hausdorffization is the different quotient

$$\mathrm{HH}_0^{\mathrm{cont}}(\mathcal{A})_{\mathrm{H}} = \mathcal{A}/\overline{[\mathcal{A}, \mathcal{A}]_{\mathrm{cont}}}.$$

Theorem 1.1 computes the former group (and its filtered associated graded); it does not prove that the commutator image is closed. Continuous traces factor through the Hausdorffization.

For the canonical b-groupoid, the continuous-trace theorem of [1] identifies those traces with the top Laurent residue classes. If $n \geq 2$, $X$ is connected, and $\partial X$ is connected, the boundary cosphere $Z$ is connected and carries its canonical contact/boundary orientation, so $H^{2n-2}(Z) \cong \mathbb{C}$. The trace theorem, not an unproved closedness claim, therefore shows that the space of continuous traces is one-dimensional.

If $n \geq 2$, $X$ is connected, and $\partial X$ is connected, then the resulting trace is the b-residue trace up to scale. With several boundary components, the same theorem gives one boundary-face residue for each connected component of $Z$.

### 11.3 A product cylinder

Let $X = [0, 1] \times N$, where $N$ is closed of dimension $n - 1$, and put

$$W = S(\mathbb{R} \oplus T^*N).$$

Then $Y \simeq [0, 1] \times W$ and $Z = W \sqcup W$. Therefore

$$\mathrm{HH}_k^{\mathrm{cont}}(\mathcal{A}_{b,L}) \cong H^{2n-k}(W) \oplus H^{2n-k-1}(W)^{\oplus 3} \oplus H^{2n-k-2}(W)^{\oplus 2}.$$

This example separates the order-circle contribution from the two boundary faces.

## 12 The unreduced operator algebra

The unreduced operator algebra fits into

$$0 \longrightarrow \Psi_{b,L}^{-\infty}(X; E) \longrightarrow \Psi_{b,L}^{\mathbb{Z}}(X; E) \longrightarrow \mathcal{A}_{b,L}(X; E) \longrightarrow 0. \tag{17}$$

The answer for $\Psi^{\mathbb{Z}}_{b,L}$ is not obtained by simply reusing (1). It depends on the topology placed on the residual ideal and on whether kernels are required to vanish rapidly at the side faces, merely have polyhomogeneous expansions, or are completed in an operator norm.

For an H-unital residual ideal, excision gives the long exact sequence

$$\cdots \to \mathrm{HH}_k(\Psi^{-\infty}_{b,L}) \to \mathrm{HH}_k(\Psi^{\mathbb{Z}}_{b,L}) \to \mathrm{HH}_k(\mathcal{A}_{b,L}) \xrightarrow{\partial} \mathrm{HH}_{k-1}(\Psi^{-\infty}_{b,L}) \to \cdots .$$

Melrose and Nistor compute the analogous two-filtration sequence in the cusp calculus and explain how the same mechanism applies to the b-calculus [8].

Here H-unitality means that the completed bar complex of the nonunital residual ideal is acyclic; this is the hypothesis that prevents an additional excision defect. Continuous excision and H-unitality are substantial theorems of Wodzicki type [12]. For the standard rapidly decreasing nuclear smoothing algebra the hypothesis is known, but it is not proved by a one-line finite-rank approximation. Allowing polyhomogeneous side-face terms changes both topology and residual homology, so H-unitality must be verified anew for the chosen completion.

The Hochschild connecting morphism can be seen on a symbol cycle. Choose operator lifts of its tensor factors, apply the Hochschild boundary upstairs, and observe that the result lies in the residual ideal because its image vanishes in the quotient. The resulting residual homology class is independent of the lifts. This Hochschild map is not literally the Fredholm boundary map in $K$-theory. Compatibility appears only after applying the Chern character to cyclic homology and using excision in both theories. Boundary logarithmic classes then pair with regularized cyclic cocycles; the eta term is an analytic realization of such a pairing, not an element produced by the Hochschild boundary alone.

Thus (1) is the complete answer for Laurent complete symbols, while (17), together with the homology of the chosen residual ideal, is the correct computation for a specified unreduced operator algebra.

## 13 Relation to index theory

An elliptic b-operator determines an invertible matrix over the complete-symbol algebra. A Fredholm index additionally requires invertibility of its normal family. Its $K_1$-class has a Chern character in cyclic homology, which can pair with cyclic cocycles arising from the Hochschild classes above. In this qualified sense, the ordinary residue class supplies the interior symbolic term and a boundary logarithmic cocycle supplies the indicial correction. The operator-extension boundary map is the Fredholm index only under full ellipticity.

The work of Liu and Ma on differential $K$-theory and eta-invariant localization supplies a geometric refinement of this pairing: differential forms and eta data refine the topological $K$-class. It does not replace the Hochschild spectral-sequence calculation [6]. The logical division is therefore

Melrose–Nistor and Benameur–Nistor: algebraic homology,

Liu–Ma: differential-$K$ and eta localization of index data.

Mazzeo–Melrose's fibred-boundary calculus gives complementary analytic background for normal families, mapping properties, and Fredholm criteria [10]. It does not replace the Benameur–Nistor topologically filtered Hochschild theorem. This distinction is useful when transporting the present computation to edge or fibred-cusp settings.

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